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11 Write a program for the task of Credit Score Classification

Program:

```
# Credit Score Classification
```

```
# Using Decision Tree
```

```
from sklearn.model_selection import train_test_split
```

```
from sklearn.tree import DecisionTreeClassifier
```

```
from sklearn.metrics import accuracy_score
```

```
# Dataset
```

```
# [Age, Income, Loan Amount, Number of Loans]
```

```
X = [
```

```
    [22, 20000, 5000, 1],
```

```
    [25, 25000, 7000, 2],
```

```
    [28, 30000, 8000, 2],
```

```
    [30, 35000, 10000, 2],
```

```
    [35, 45000, 12000, 3],
```

```
    [40, 50000, 15000, 3],
```

```
    [45, 60000, 18000, 4],
```

```
    [50, 70000, 20000, 4],
```

```
    [23, 22000, 6000, 1],
```

```
    [27, 28000, 9000, 2],
```

```
    [33, 40000, 11000, 3],
```

```
    [38, 48000, 14000, 3],
```

```
[43, 55000, 17000, 4],  
[48, 65000, 19000, 4],  
[52, 75000, 22000, 5]  
]  
  
# Credit categories  
# 0 = Poor  
# 1 = Standard  
# 2 = Good  
  
y = [  
    0, 0, 0, 1, 1,  
    1, 2, 2, 0, 0,  
    1, 1, 2, 2, 2  
]  
  
# Split data  
X_train, X_test, y_train, y_test = train_test_split(  
    X, y,  
    test_size=0.3,  
    random_state=42  
)  
  
# Create model  
model = DecisionTreeClassifier(random_state=42)  
  
# Train model  
model.fit(X_train, y_train)  
  
# Predict
```

```

y_pred = model.predict(X_test)

# Accuracy
accuracy = accuracy_score(y_test, y_pred)

print("=====")
print(" CREDIT SCORE CLASSIFICATION")
print("=====")

print("\nActual Values :", y_test)
print("Predicted Values :", y_pred)

print("\nAccuracy:", accuracy)
print("Accuracy Percentage:", accuracy * 100, "%")

# Predict a new customer
new_customer = [[32, 40000, 11000, 3]]

prediction = model.predict(new_customer)

print("\nNew Customer Details")
print("-----")
print("Age          : 32")
print("Annual Income  : 40000")
print("Loan Amount    : 11000")
print("Number of Loans : 3")

if prediction[0] == 0:
    credit_score = "Poor"
elif prediction[0] == 1:

```

```
        credit_score = "Standard"
else:
    credit_score = "Good"

print("\nPredicted Credit Score:", credit_score)
```

output:

```
=====
      CREDIT SCORE CLASSIFICATION
=====

Actual Values      : [0, 1, 0, 2, 1]
Predicted Values   : [0 1 0 2 2]

Accuracy: 0.8
Accuracy Percentage: 80.0 %

New Customer Details
-----
Age                : 32
Annual Income      : 40000
Loan Amount        : 11000
Number of Loans    : 3

Predicted Credit Score: Standard
```